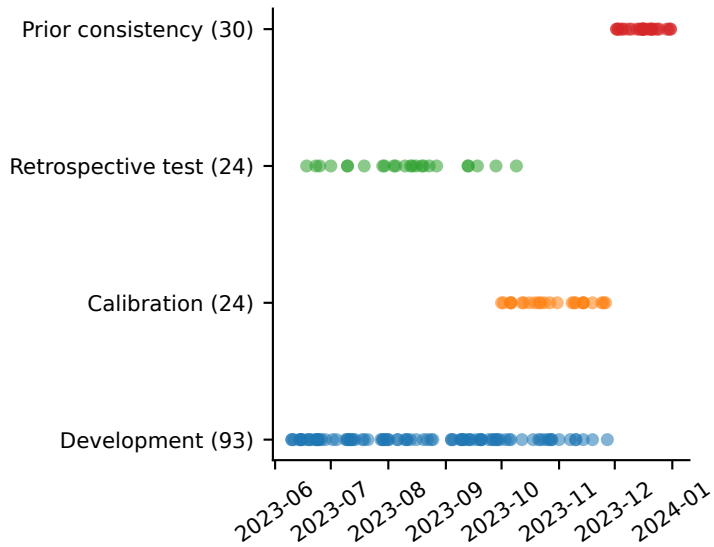


(a) Whole-earthquake roles



(b) Historical waveform timing

